

Object Detection Fundamentals

An educational guide to object detection, evaluation, and modern detection pipelines.

Learning goals

- Understand the difference between image classification, localization, and detection.
- Understand bounding boxes, confidence scores, and class labels.
- Understand IoU and Non-Maximum Suppression.
- Understand precision, recall, and mAP.
- Understand the basic idea behind YOLO-style real-time detection.

What is object detection?

Object detection answers two questions: **what** objects are present and **where** they are located. A detector usually outputs a class label, a confidence score, and a bounding box for each detected object.

Task	Output
Classification	One or more labels for an image.
Object localization	A class plus location for a main object.
Object detection	Multiple objects, each with class and location.

1. Bounding Boxes and Detection Outputs

A bounding box is a rectangle surrounding an object. It may be represented by coordinates such as `x_min`, `y_min`, `x_max`, `y_max`, or by center coordinates plus width and height.

Typical prediction

Component	Example
Class	person
Confidence	0.91
Bounding box	(120, 80, 260, 410)

Confidence score

The confidence score expresses how strongly the model believes a prediction is valid according to its scoring system. It should not automatically be interpreted as a perfectly calibrated probability.

Thresholding

A system may discard predictions below a chosen confidence threshold. A higher threshold usually produces fewer predictions, which can reduce false positives but may also remove true objects.

Real-world example

A traffic camera might detect cars, buses, motorcycles, bicycles, and pedestrians. Each detection has a location so downstream software can count objects or monitor zones.

2. IoU — Intersection over Union

Intersection over Union (IoU) measures the overlap between two bounding boxes.

IoU = Area of Intersection / Area of Union

IoU ranges from 0 to 1 for ordinary non-negative box areas. Zero means no overlap; one means the boxes are identical.

Why IoU matters

A predicted box should not be considered correct merely because it overlaps an object. IoU gives a quantitative way to decide whether the predicted location is sufficiently close to a ground-truth box.

Situation	Typical interpretation
IoU = 0	No overlap
Low IoU	Poor localization
High IoU	Strong overlap
IoU = 1	Exact same box

Evaluation rule

An evaluation protocol may declare a prediction a true positive when its IoU with an appropriate ground-truth object exceeds a selected threshold, while also enforcing class agreement and matching rules.

3. Non-Maximum Suppression

Object detectors can produce several overlapping boxes around the same object. Non-Maximum Suppression (NMS) reduces duplicate detections.

Basic NMS process

- Sort candidate boxes by confidence score.
- Select the highest-confidence box.
- Compare it with remaining boxes.
- Suppress boxes whose overlap with the selected box exceeds an IoU threshold.
- Repeat with remaining boxes.

Example

Suppose three boxes detect the same car with confidence scores 0.93, 0.88, and 0.72. If the boxes overlap heavily, NMS may keep the 0.93 prediction and remove the duplicates.

Trade-off

If the NMS IoU threshold is too low, valid nearby objects may be suppressed. If it is too high, duplicate predictions may remain.

Modern variants

Many detection systems use improved variants such as class-aware NMS, class-agnostic NMS, Soft-NMS, or model-specific post-processing.

4. Precision and Recall

Detection quality involves a trade-off between finding objects and avoiding incorrect detections.

Metric	Formula	Meaning
Precision	$TP / (TP + FP)$	Of predicted positives, how many are correct?
Recall	$TP / (TP + FN)$	Of actual positives, how many were found?

False positive

The model predicts an object when the required object is not actually present or the prediction does not satisfy the evaluation matching criteria.

False negative

An actual object exists, but the detector fails to produce a valid matching prediction.

Why thresholds matter

Changing the confidence threshold can change precision and recall. A system used for safety monitoring may prioritize recall, while a system that triggers expensive downstream actions may prioritize precision.

Important

Precision and recall should be evaluated together rather than using accuracy alone. Object detection has many possible background locations and is not a simple one-label-per-image problem.

5. Average Precision and mAP

Average Precision (AP) summarizes precision-recall performance for a class under a defined evaluation protocol. Mean Average Precision (mAP) averages AP across classes and, depending on the benchmark, may also average across multiple IoU thresholds.

Why mAP is useful

A single confidence threshold can hide the behavior of a detector. Precision-recall curves evaluate performance across thresholds, while AP summarizes the curve according to the chosen convention.

Term	Meaning
AP	Average Precision for a particular class/evaluation setting.
mAP	Mean of AP values across classes and/or specified IoU settings.
mAP@0.5	Common shorthand for evaluation at IoU 0.5 in some workflows.
mAP@0.5:0.95	Common shorthand for averaging across IoU thresholds from 0.50 to 0.95 at specified increments.

Do not compare blindly

Metrics are only comparable when the dataset, class definitions, IoU thresholds, matching rules, and evaluation implementation are aligned.

6. YOLO-Style Detection

YOLO, meaning “You Only Look Once,” is a family of object detection methods designed around efficient end-to-end prediction. Different YOLO versions have different architectures and training strategies, so the name should not be treated as one fixed algorithm.

High-level pipeline

- Input image is processed by a neural network.
- The model produces predictions for object locations and classes.
- Confidence filtering and post-processing may remove weak or duplicate predictions.
- Remaining detections are returned to the application.

Why YOLO became popular

- Fast inference suitable for real-time or near-real-time applications.
- Single-stage detection designs can be efficient.
- Large ecosystem of pretrained models and tools.
- Useful for applications such as traffic monitoring, manufacturing inspection, and robotics.

Trade-offs

Speed, accuracy, memory use, input resolution, and hardware cost must be balanced. A model that is excellent on a benchmark may not be the best choice for a specific deployment environment.

7. Object Detection in a Real Application

Example: warehouse safety monitoring

- Camera captures an image or video frame.
- Detector identifies workers, forklifts, and restricted-area objects.
- A zone-checking rule determines whether a detected person is inside a restricted region.
- An event is logged when conditions are satisfied.
- A dashboard displays counts, trends, and alerts.

Important system components

Component	Responsibility
Camera / input	Provides images or video frames.
Detection model	Finds objects and locations.
Post-processing	Filters and removes duplicate predictions.
Business rules	Converts detections into application events.
Database	Stores events and metrics.
Dashboard	Shows trends and operational information.

Final takeaway

Object detection is more than drawing boxes. A reliable system requires data quality, model selection, threshold tuning, evaluation, post-processing, and application-level logic. These concepts also make good RAG test material because they contain definitions, procedures, formulas, and examples.